

(9)

CSA0711

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Code of Conduct:

I Dinesh Bhavadi (192572028) Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
 I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: D. Bhavadi

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5M
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	2	2	2	2	2
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	1	1	1	1

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6

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26/30

26/28

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Smart campus subnet Allocation:-

Basic Network Block: 192.168.100.0/24
Subnet Allocations:-

- i) School of Computing: 125
- ii) School of Electronics: 126
- iii) School of Mechanical Engineering: 127

i) School of Computing (125):

→ Subnet Mask: 255.255.255.128

→ Block size = 128 IP addresses

→ Network Address: 192.168.100.0

→ Valid Host IP Range: 192.168.100.1 - 192.168.100.126

→ Usable Host Addresses = 126 hosts

ii) School of Electronics (126):

→ Subnet Mask: 255.255.255.192

→ Block size = 64 IP addresses

→ Network Address = 192.168.100.128

→ Valid Host IP Range: 192.168.100.129 - 192.168.100.190

→ Usable Host Address: 62 hosts

iii) school of Mechanical Engineering (127)

→ Subnet Mask: 255.255.255.224

→ Block size = 32 IP addresses

→ Network addresses: 192.168.100.192

→ IP Range: 192.168.100.192 - 192.168.100.223

→ Broadcast Addresses: 192.168.100.223

→ Valid Host IP Range: 192.168.100.193 - 192.168.100.222

→ usable Host Addresses: 30 hosts

Remaining unused IP address Range:-

Allocated IPs: 192.168.100.0 to 192.168.100.223

~~use~~ unused Network Range: 192.168.100.224 - 192.168.100.255

Total unused Addresses: 256 - 224 = 32

Company Network Subnetting

Basic Network Block: 10.10.0.0/16

Subnet Allocations:-

→ Software Development: 124